

2017 HSC Mathematics Extension 2

Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

Question	Answer
1	C
2	B
3	D
4	C
5	B
6	A
7	A
8	B
9	C
10	B

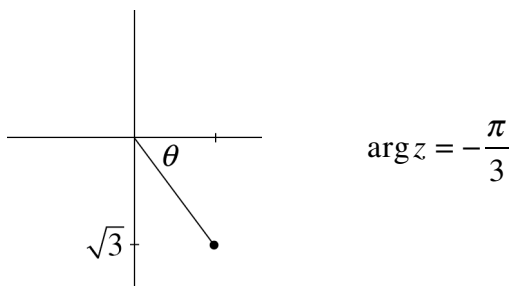
Section II

Question 11 (a) (i)

Criteria	Marks
• Provides correct value	1

Sample answer:

$$z = 1 - i\sqrt{3}$$



Question 11 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds argument of w , or equivalent merit	1

Sample answer:

$$w = 1 + i$$

$$\arg w = \frac{\pi}{4}$$

$$\arg\left(\frac{z}{w}\right) = \arg z - \arg w$$

$$= \left(-\frac{\pi}{3}\right) - \left(\frac{\pi}{4}\right)$$

$$= \frac{-4\pi - 3\pi}{12}$$

$$= \frac{-7\pi}{12}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Finds the slope of the asymptote, or equivalent merit	1

Sample answer:

$$\frac{x^2}{12} - \frac{y^2}{4} = 1$$

$$a^2 = 12 \quad \therefore a = 2\sqrt{3}$$

$$b^2 = 4 \quad \therefore b = 2$$

$$\tan \alpha = \frac{b}{a} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

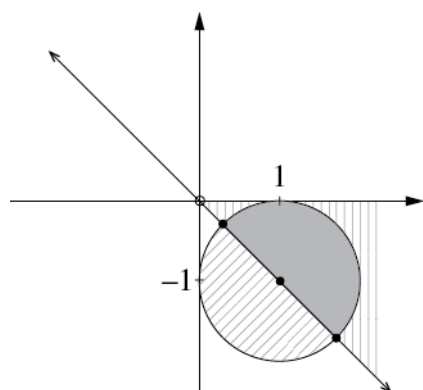
$$\alpha = \frac{\pi}{6}$$

Question 11 (c)

Criteria	Marks
• Provides correct sketch	2
• Sketches one correct region, or equivalent merit	1

Sample answer:

$$-\frac{\pi}{4} \leq \arg z \leq 0 \quad \text{and} \quad |z - (1 - i)| \leq 1$$



Question 11 (d)

Criteria	Marks
• Provides correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Correctly deals with $d\theta$, or equivalent merit	1

Sample answer:

$$t = \tan \frac{\theta}{2}$$

$$\int_0^{\frac{2\pi}{3}} \frac{1}{1 + \cos \theta} \times d\theta = \int_0^{\sqrt{3}} \frac{\frac{2dt}{1+t^2}}{\frac{1+t^2}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^{\sqrt{3}} \frac{2dt}{2}$$

$$= \int_0^{\sqrt{3}} 1 \times dt$$

$$= t \Big|_0^{\sqrt{3}}$$

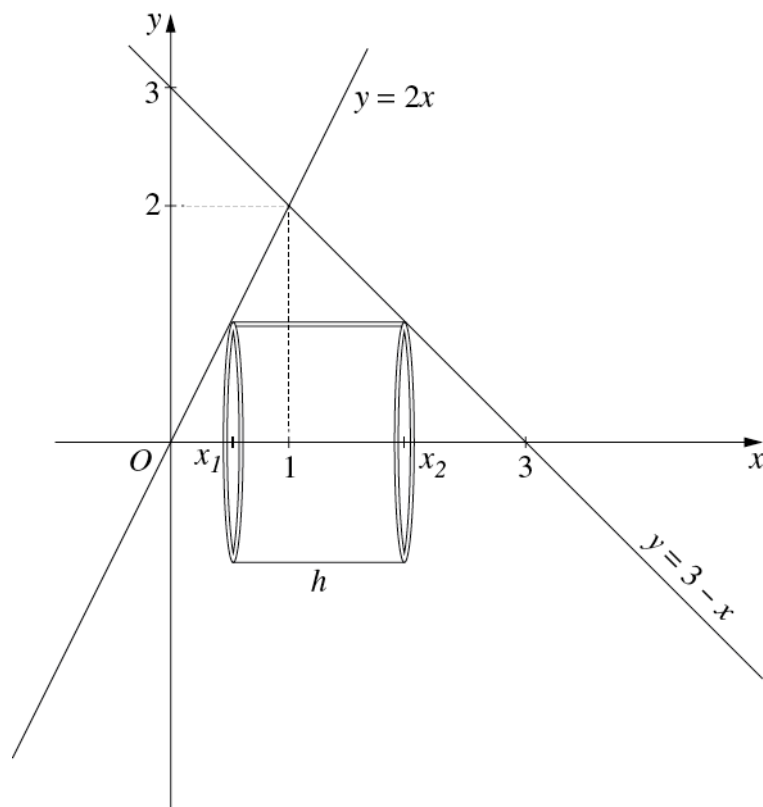
$$= \sqrt{3} - 0$$

$$= \sqrt{3}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	2
• Finds the height of the shells in terms of y , or equivalent merit	1

Sample answer:



$$\begin{aligned}
 r &= y \\
 h &= x_2 - x_1 \\
 &= (3 - y) - \left(\frac{y}{2}\right) \\
 &= 3 - \frac{3}{2}y
 \end{aligned}$$

$$\begin{aligned}
 V &= 2\pi \int_0^2 rh \, dy \\
 &= 2\pi \int_0^2 y \times \left(3 - \frac{3}{2}y\right) \times dy
 \end{aligned}$$

Question 11 (f)

Criteria	Marks
• Provides correct solution	3
• Attempts to use a double angle result, or equivalent merit	2
• Obtains correct integrand in terms of θ , or equivalent merit	1

Sample answer:

$$x = \sin^2 \theta$$

$$dx = 2\sin\theta\cos\theta d\theta$$

$$\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} \times dx = \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2\theta}{1-\sin^2\theta}} \times 2\sin\theta\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin\theta}{\cos\theta} \times 2\sin\theta\cos\theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \sin^2\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} (1 - \cos 2\theta) d\theta$$

$$= \theta - \frac{1}{2}\sin 2\theta \Bigg|_0^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{4} - \frac{1}{2}\sin \frac{\pi}{2} \right) - (0)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$f(x) = \frac{e^x - 1}{e^x + 1} \quad u = e^x - 1$$

$$u = e^x$$

$$v = e^x + 1$$

$$v = e^x$$

$$f'(x) = \frac{(e^x + 1)e^x - (e^x - 1)e^x}{(e^x + 1)^2}$$

$$= \frac{e^{2x} + e^x - e^{2x} + e^x}{(e^x + 1)^2}$$

$$= \frac{2e^x}{(e^x + 1)^2}$$

$$f'(x) > 0 \text{ for all } x \text{ as } 2e^x > 0 \text{ and } (e^x + 1)^2 > 0$$

$\therefore f(x)$ is increasing for all x .

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}
 f(-x) &= \frac{e^{-x} - 1}{e^{-x} + 1} \\
 &= \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \\
 &= \frac{\frac{1 - e^x}{e^x}}{\frac{1 + e^x}{e^x}} \\
 &= \frac{1 - e^x}{1 + e^x} \\
 &= \frac{-(-1 + e^x)}{e^x + 1} \\
 &= -\frac{(e^x - 1)}{e^x + 1} \\
 &= -[f(x)]
 \end{aligned}$$

$$\therefore f(-x) = -f(x)$$

the function is odd.

Question 12 (a) (iii)

Criteria	Marks
• Provides correct description	1

Sample answer:

$$\lim_{x \rightarrow \infty} \frac{e^x - 1}{e^x + 1} = 1$$

ie $x \rightarrow \infty$, $f(x) \rightarrow 1$ (from below)

Question 12 (a) (iv)

Criteria	Marks
• Provides correct sketch	1

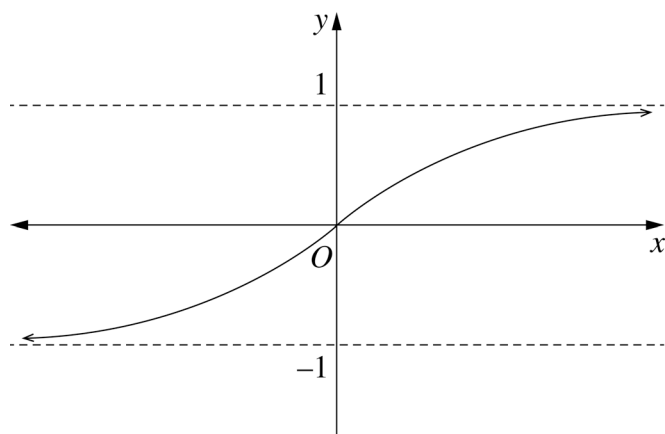
Sample answer:

$$f(0) = \frac{e^0 - 1}{e^0 + 1}$$

$$= \frac{1 - 1}{1 + 1}$$

$$= 0$$

$\therefore (0, 0)$ lies on the curve.



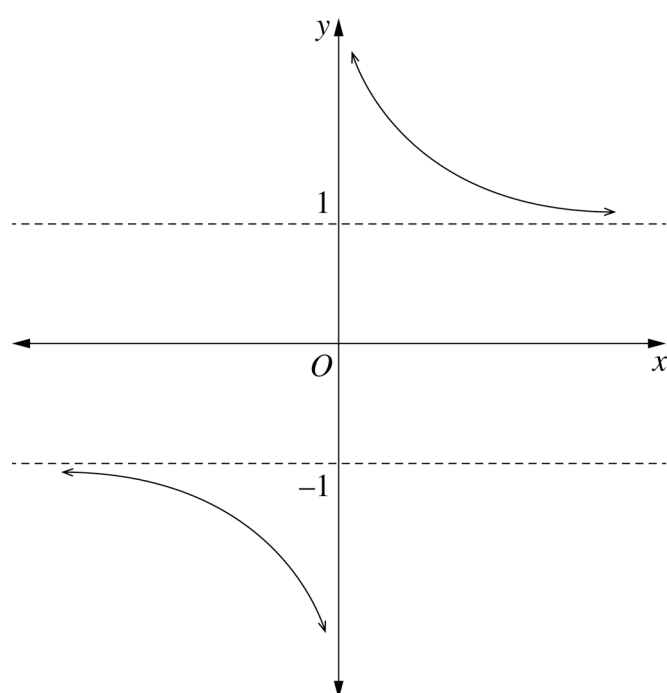
Question 12 (a) (v)

Criteria	Marks
• Provides correct sketch	1

Sample answer:

$$y = \frac{e^x + 1}{e^x - 1}$$

$$y = \frac{1}{f(x)}$$



Question 12 (b)

Criteria	Marks
• Provides correct solution	3
• Correctly evaluates the discriminant, or equivalent merit	2
• Uses quadratic formula, or equivalent merit	1

Sample answer:

$$z^2 + (2 + 3i)z + (1 + 3i) = 0$$

$$a = 1$$

$$b = 2 + 3i$$

$$c = 1 + 3i$$

$$z = \frac{-(2 + 3i) \pm \sqrt{(2 + 3i)^2 - 4(1)(1 + 3i)}}{2(1)}$$

$$= \frac{-(2 + 3i) \pm \sqrt{4 + 12i + 9i^2 - 4 - 12i}}{2}$$

$$= \frac{-(2 + 3i) \pm \sqrt{9i^2}}{2}$$

$$= \frac{-(2 + 3i) \pm 3i}{2}$$

$$z = \frac{-2 - 3i + 3i}{2} \quad \text{and} \quad z = \frac{-2 - 3i - 3i}{2}$$

$$z = \frac{-2}{2} \quad \text{and} \quad z = \frac{-2 - 6i}{2}$$

$$z = -1 \quad z = -1 - 3i$$

Question 12 (c)

Criteria	Marks
• Provides correct primitive	3
• Correctly uses parts, or equivalent merit	2
• Attempts to use parts, or equivalent merit	1

Sample answer:

$$I = \int x \tan^{-1} x \, dx$$

Using integration by parts,

$$u = \tan^{-1} x \quad dv = x dx$$

$$du = \frac{1}{1+x^2} dx \quad v = \frac{x^2}{2}$$

$$\begin{aligned}
 I &= \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \left(\frac{1}{1+x^2} \right) dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2 + 1 - 1}{x^2 + 1} dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) dx \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[x - \tan^{-1} x \right] + c \\
 &= \frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{\tan^{-1} x}{2} + c \\
 &= \frac{1}{2} (x^2 \tan^{-1} x - x + \tan^{-1} x) + c
 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct solution	2
• Recognises that $P(x) = (x - \alpha)^2 q(x)$, or equivalent merit	1

Sample answer:

$$\text{Let } P(x) = (x - \alpha)^2 \times Q(x)$$

$$P'(x) = 2(x - \alpha)' \times Q(x) + (x - \alpha)^2 \times Q'(x)$$

$$= (x - \alpha)[2 \times Q(x) + (x - \alpha) \times Q'(x)]$$

$$P(\alpha) = (\alpha - \alpha)^2 \times Q(\alpha)$$

$$= 0$$

$$P'(\alpha) = (\alpha - \alpha)[2 \times Q(\alpha) + (\alpha - \alpha) \times Q'(\alpha)]$$

$$= 0$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempt to solve $P'(x) = 0$, or equivalent merit	1

Sample answer:

$$P(x) = x^4 - 3x^3 + x^2 + 4$$

$$P'(x) = 4x^3 - 9x^2 + 2x$$

$$= x(4x^2 - 9x + 2)$$

$$= x(4x - 1)(x - 2)$$

Roots of $P'(x)$ are $x = 0, \frac{1}{4}, 2$

$$P(0) = 4$$

$$P\left(\frac{1}{4}\right) = \frac{1029}{256}$$

$$P(2) = 0$$

$\therefore (x - 2)^2$ is a factor of $P(x)$

$$\alpha = 2$$

Question 13 (a)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\text{As } (\sqrt{r} - \sqrt{s})^2 \geq 0$$

$$r - 2\sqrt{rs} + s \geq 0$$

$$\therefore \frac{r+s}{2} \geq \sqrt{rs}$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct proof	2
• Attempts to use sum of the threewise products of roots, or equivalent merit	1

Sample answer:

$$-a = \text{sum of roots} = \alpha + \frac{1}{\alpha} + \beta + \frac{1}{\beta}$$

$$-c = \text{sum of products of 3 roots}$$

$$= \alpha \frac{1}{\alpha} \beta + \alpha \frac{1}{\alpha} \frac{1}{\beta} + \alpha \beta \frac{1}{\beta} + \frac{1}{\alpha} \beta \frac{1}{\beta}$$

$$= \beta + \frac{1}{\beta} + \alpha + \frac{1}{\alpha}$$

$$= -a$$

$$\therefore a = c$$

OR

The equation whose roots are reciprocals of the roots is found by replacing x by $\frac{1}{x}$.

$$\frac{1}{x^4} + a \frac{1}{x^3} + b \frac{1}{x^2} + c \frac{1}{x} + d$$

$$\text{ie } 1 + ax + bx^2 + cx^3 + dx^4$$

Since the reciprocals give the same set of roots we can match the coefficients so that $d = 1$ and $c = a$.

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses sum of the roots in pairs, or equivalent merit	1

Sample answer:

b = sum of pairs of roots

$$= \alpha \frac{1}{\alpha} + \alpha\beta + \alpha \frac{1}{\beta} + \beta \frac{1}{\alpha} + \frac{1}{\alpha} \frac{1}{\beta} + \beta \frac{1}{\beta}$$

$$= 1 + \alpha\beta + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \frac{1}{\alpha\beta} + 1$$

$$= 1 + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \alpha\beta + \frac{1}{\alpha\beta}$$

Now $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} \geq \frac{2\sqrt{\alpha^2\beta^2}}{\alpha\beta}$ from 13 part (a)

$$\therefore \frac{\alpha}{\beta} + \frac{\beta}{\alpha} \geq 2 \quad \text{and similarly} \quad \alpha\beta + \frac{1}{\alpha\beta} \geq 2$$

$$\therefore b \geq 2 + 2 + 2 = 6$$

Question 13 (c)

Criteria	Marks
• Provides correct solution	4
• Obtains correct expression for x in terms of v , k and g , or equivalent merit	3
• Separates the variables, or equivalent merit	2
• Rewrites acceleration as $v \frac{dv}{dx}$, or equivalent merit	1

Sample answer:

$$\ddot{x} = v \frac{dv}{dx} = -g - kv^2$$

$$\therefore \frac{dv}{dx} = -\frac{g + kv^2}{v}$$

$$\frac{dx}{dv} = -\frac{v}{g + kv^2}$$

$$\therefore x = -\int \frac{v}{g + kv^2} dv$$

$$= -\frac{1}{2k} \log_e(g + kv^2) + c$$

when $x = 0$ $v = \frac{1}{2} \sqrt{\frac{g}{k}}$ so that $v^2 = \frac{g}{4k}$

hence $0 = -\frac{1}{2k} \log_e\left(g + \frac{kg}{4k}\right) + c$

$$\therefore c = \frac{1}{2k} \log_e\left(\frac{5g}{4}\right)$$

$$\therefore x = \frac{1}{2k} \log_e\left(\frac{5g}{4}\right) - \frac{1}{2k} \log_e(g + kv^2)$$

Maximum height occurs when $v = 0$

$$\therefore H = \frac{1}{2k} \left[\log_e\left(\frac{5g}{4}\right) - \log_e(g) \right]$$

$$= \frac{1}{2k} \log_e\left(\frac{5}{4}\right)$$

Question 13 (d)

Criteria	Marks
• Provides correct solution	4
• Obtains an integral for the volume, or equivalent merit	3
• Finds the area of the cross-section in terms of x , or equivalent merit	2
• Finds the height of $\triangle PQR$ in terms of x , or equivalent merit	1

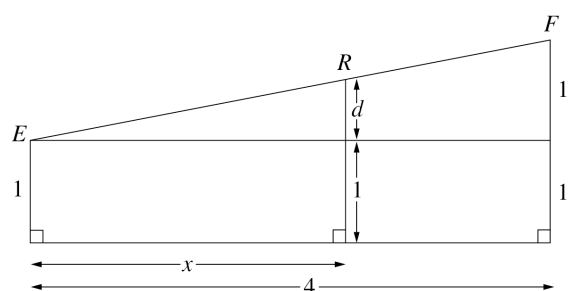
Sample answer:

The line BC has equation $y = \left(\frac{3-1}{4-0}\right)x + 1$

$$y = \frac{1}{2}x + 1$$

$$\therefore QP = 2\left(\frac{1}{2}x + 1\right) = x + 2$$

To find height of $\triangle QRP$



$$\therefore d = \frac{x}{4}$$

$$\therefore \text{height} = 1 + d = \frac{x}{4} + 1$$

By similar triangles $\frac{d}{x} = \frac{1}{4}$

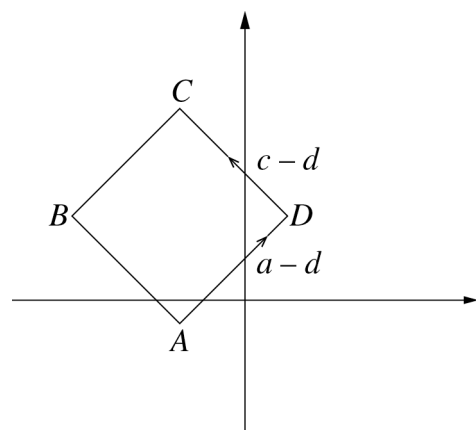
$$\begin{aligned}
 \therefore \text{Area } \triangle QRP &= \frac{1}{2}QP\left(\frac{x}{4}+1\right) \\
 &= \left(\frac{x}{2}+1\right)\left(\frac{x}{4}+1\right) \\
 &= \frac{x^2}{8} + \frac{3x}{4} + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume} &= \frac{1}{8} \int_0^4 x^2 + 6x + 8 \, dx \\
 &= \frac{1}{8} \left[\frac{x^3}{3} + 3x^2 + 8x \right]_0^4 \\
 &= \frac{1}{8} \left[\frac{64}{3} + 48 + 32 - 0 \right] \\
 &= \frac{38}{3} \text{ units}^3
 \end{aligned}$$

Question 13 (e)

Criteria	Marks
• Provides correct solution	2
• Obtains $c - d = i(d - a)$, or equivalent merit	1

Sample answer:



$$c - d = i(d - a)$$

(rotation by $\frac{\pi}{2}$)

$$c = d + id - ia$$

$$= (1 + i)d - ia$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct values	1

Sample answer:

Multiply both sides by $x^4 + 4$

$$16 = (A + 2x)(x^2 - 2x + 2) + (B - 2x)(x^2 + 2x + 2)$$

$$\text{at } x = 0 \quad 16 = 2A + 2B \quad A + B = 8 \quad \text{--- ①}$$

$$\begin{aligned} \text{at } x = 1 \quad 16 &= (A + 2) + (B - 2)5 \\ 16 &= A + 2 + 5B - 10 \quad A + 5B = 24 \quad \text{--- ②} \end{aligned}$$

Subtract ② – ①

$$4B = 16$$

$$B = 4$$

$$A = 4$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains correct primitive, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int_0^m \frac{16}{x^4 + 4} dx \\
 &= \int_0^m \frac{4 + 2x}{x^2 + 2x + 2} + \int_0^m \frac{4 - 2x}{x^2 - 2x + 2} dx \\
 &= \int_0^m \frac{2}{x^2 + 2x + 2} + \frac{2x + 2}{x^2 + 2x + 2} dx + \int_0^m \frac{2}{x^2 - 2x + 2} - \frac{2x - 2}{x^2 - 2x + 2} dx \\
 &= \int_0^m \frac{2}{1 + (x + 1)^2} dx + \left[\ln(x^2 + 2x + 2) \right]_0^m + \int_0^m \frac{2}{1 + (x - 1)^2} dx - \left[\ln(x^2 - 2x + 2) \right]_0^m \\
 &= \left[2 \tan^{-1}(x + 1) \right]_0^m + \ln(m^2 + 2m + 2) - \log 2 + \left[2 \tan^{-1}(x - 1) \right]_0^m - \ln(m^2 - 2m + 2) + \log 2 \\
 &= 2 \tan^{-1}(m + 1) - \frac{\pi}{2} + \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m - 1) + \frac{\pi}{2} \\
 &= \ln\left(\frac{m^2 + 2m + 2}{m^2 - 2m + 2}\right) + 2 \tan^{-1}(m + 1) + 2 \tan^{-1}(m - 1)
 \end{aligned}$$

Question 14 (a) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\int_0^m \frac{16}{x^4 + 4} dx = \ln \left(\frac{1 + \frac{2}{m} + \frac{2}{m^2}}{1 - \frac{2}{m} + \frac{2}{m^2}} \right) + 2 \tan^{-1}(m+1) + 2 \tan^{-1}(m-1)$$

as $m \rightarrow \infty$

$$\rightarrow \ln(1) + 2\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right) = 2\pi$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Construct DB

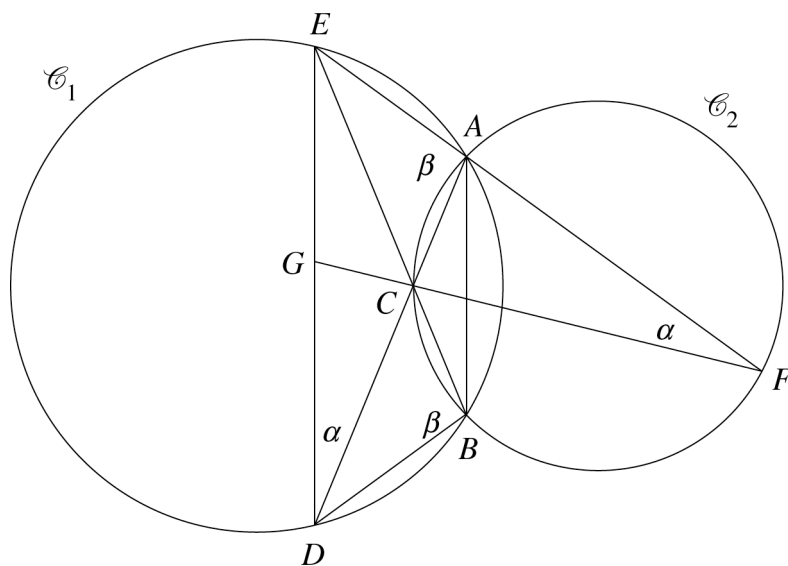
$\angle EAD = \angle EBD$, as they stand on the same arc, ED of C_1 .

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Construct AB



$\angle EDA = \angle EBA$ angles standing on arc EA of C_1 .

$\angle EBA = \angle CBA$
 $= \angle CFA$ angles standing on arc AC in other circle C_2 .

so $\angle EDA = \angle EBA = \angle CFA = \angle AFC$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Makes substantial progress, or equivalent merit	2
• Attempts to find an exterior angle, or equivalent merit	1

Sample answer:

Let $\angle AFC = \angle EDA = \alpha$
 $\angle EAD = \angle EBD = \beta$

In $\triangle EAD$, $\angle AED = \pi - \alpha - \beta$

In $\triangle EFG$, $\angle EGF = \pi - (\alpha + (\pi - \alpha - \beta))$
 $= \beta$

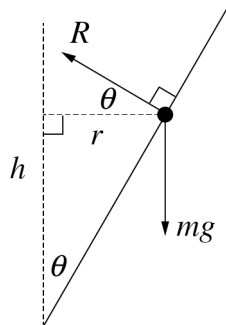
so $\angle EGF = \angle EBD = \beta$

Exterior angle (EGF) equals opposite interior angle (EBD) and so $BCGD$ is a cyclic quadrilateral.

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains one correct component of R , or equivalent merit	1

Sample answer:



$$R \sin \theta = mg$$

$$R \cos \theta = mrw^2$$

Dividing

$$\tan \theta = \frac{g}{rw^2}$$

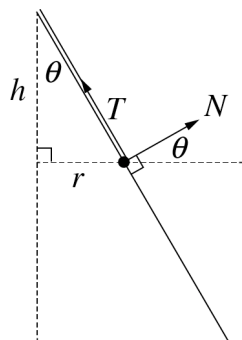
$$\frac{r}{h} = \frac{g}{rw^2}$$

$$w^2 = \frac{gh}{r^2}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly resolves N , or equivalent merit	1

Sample answer:



Horizontal:

$$T \sin \theta - N \cos \theta = mrw^2$$

$$T \sin \theta \cos \theta - N \cos^2 \theta = mrw^2 \cos \theta \quad \text{--- ①}$$

Vertical:

$$T \cos \theta + N \sin \theta = mg$$

$$T \sin \theta \cos \theta + N \sin^2 \theta = mg \sin \theta \quad \text{--- ②}$$

$$\text{②} - \text{①} \quad N(\sin^2 \theta + \cos^2 \theta) = mg \sin \theta - mrw^2 \cos \theta$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Observes $N \geq 0$, or equivalent merit	1

Sample answer:

As particle 2 moves in a circle of radius r it must have

$$N \geq 0$$

$$mg\left(\sin\theta - \frac{h}{r}\cos\theta\right) \geq 0$$

$$\sin\theta - \cot\theta\cos\theta \geq 0$$

$$\sin\theta \geq \cot\theta\cos\theta$$

$$\tan\theta \geq \cot\theta$$

In first quad so

$$\theta \geq \frac{\pi}{4}$$

Question 15 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 I_1 &= \int_0^1 x^1 \sqrt{1-x^2} \, dx \\
 &= \frac{1}{2} \int_0^1 -2x \sqrt{1-x^2} \, dx \\
 &= -\frac{1}{2} \left[\frac{2}{3} (1-x^2)^{\frac{3}{2}} \right]_0^1 \\
 &= -\frac{1}{3} (0-1) \\
 &= \frac{1}{3}
 \end{aligned}$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct solution	3
• Correctly uses integration by parts, simplifying where possible, or equivalent merit	2
• Attempts to use integration by parts, or equivalent merit	1

Sample answer:

$$I_n = \int_0^1 x^n \sqrt{1-x^2} dx$$

Let $u = x^{n-1}$ $dv = x\sqrt{1-x^2} dx$

$$du = (n-1)x^{n-2}dx \quad v = -\frac{1}{3}(1-x^2)^{\frac{3}{2}}$$

$$\therefore I_n = \left[x^{n-1} \times \frac{-1}{3}(1-x^2)^{\frac{3}{2}} \right]_0^1 + \frac{1}{3} \int_0^1 (1-x^2)^{\frac{3}{2}} (n-1)x^{n-2} dx$$

$$= \frac{(n-1)}{3} \int_0^1 x^{n-2} (1-x^2) \sqrt{1-x^2} dx$$

$$= \frac{(n-1)}{3} \int_0^1 x^{n-2} \sqrt{1-x^2} dx - \frac{n-1}{3} \int_0^1 x^n \sqrt{1-x^2} dx$$

$$\therefore I_n = \frac{n-1}{3} I_{n-2} - \frac{n-1}{3} I_n$$

$$I_n \left(1 + \frac{n-1}{3} \right) = \frac{n-1}{3} I_{n-2}$$

$$I_n \left(\frac{3+n-1}{3} \right) = \frac{n-1}{3} I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n+2} I_{n-2}$$

Question 15 (a) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 I_5 &= \frac{4}{7} \times I_3 \\
 &= \frac{4}{7} \times \frac{2}{5} \times I_1 \\
 &= \frac{4}{7} \times \frac{2}{5} \times \frac{1}{3} \quad \text{from part (i)} \\
 &= \frac{8}{105}
 \end{aligned}$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Differentiates with respect to x , or equivalent merit	1

Sample answer:

$$x^{\frac{1}{2}} + y^{\frac{1}{2}} = a^{\frac{1}{2}} \quad (a \text{ constant})$$

Differentiating implicitly:

$$\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{\frac{1}{2}} \times \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}}$$

$$= -\frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$$

$$= -\sqrt{\frac{y}{x}}$$

$$\therefore \text{At } P, \text{ gradient of tangent} = -\sqrt{\frac{d}{c}}$$

and equation of tangent is

$$y - d = -\sqrt{\frac{d}{c}}(x - c)$$

$$\sqrt{c}y - \sqrt{c}d = -\sqrt{d}x + \sqrt{d}c$$

$$\therefore y\sqrt{c} + x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Obtains an expression for $OA + OB$ in terms of c and d , or equivalent merit	2
• Finds an expression for OA , or equivalent merit	1

Sample answer:

At A , $y = 0$

$$\therefore x\sqrt{d} = d\sqrt{c} + c\sqrt{d}$$

$$x = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}}$$

$$\therefore OA = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}}$$

At B , $x = 0$

$$\therefore y\sqrt{c} = d\sqrt{c} + c\sqrt{d}$$

$$\therefore OB = \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{c}}$$

$$\begin{aligned} \therefore OA + OB &= \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{d}} + \frac{d\sqrt{c} + c\sqrt{d}}{\sqrt{c}} \\ &= \sqrt{d}\sqrt{c} + c + d + \sqrt{c}\sqrt{d} \\ &= \sqrt{c}(\sqrt{d} + \sqrt{c}) + \sqrt{d}(\sqrt{d} + \sqrt{c}) \\ &= (\sqrt{c} + \sqrt{d})(\sqrt{d} + \sqrt{c}) \end{aligned}$$

But P lies on $\sqrt{x} + \sqrt{y} = \sqrt{a}$

$$\therefore \sqrt{c} + \sqrt{d} = \sqrt{a}$$

$$\begin{aligned} \therefore OA + OB &= \sqrt{a} \times \sqrt{a} \\ &= a \end{aligned}$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve the simultaneous equations, or equivalent merit	1

Sample answer:

$$\text{At } P, \quad \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1 \quad \text{and} \quad \frac{x_1^2}{c^2} - \frac{y_1^2}{d^2} = 1$$

$$\therefore \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = \frac{x_1^2}{c^2} - \frac{y_1^2}{d^2}$$

$$x_1^2 \left(\frac{1}{a^2} - \frac{1}{c^2} \right) = -y_1^2 \left(\frac{1}{d^2} + \frac{1}{b^2} \right)$$

$$x_1^2 \left(\frac{c^2 - a^2}{a^2 c^2} \right) = -y_1^2 \left(\frac{b^2 + d^2}{b^2 d^2} \right)$$

$$\therefore \frac{x_1^2}{y_1^2} = - \left(\frac{b^2 + d^2}{b^2 d^2} \right) \left(\frac{a^2 c^2}{c^2 - a^2} \right)$$

$$= \frac{a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2}$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds product of the two slopes, or equivalent merit	2
• Finds slope of tangent to ellipse, or equivalent merit	1

Sample answer:

Ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{a^2} \times \frac{b^2}{y}$$

$$\therefore \text{Gradient of tangent at } P = \frac{-x_1 b^2}{a^2 y_1}$$

Hyperbola: $\frac{x^2}{c^2} - \frac{y^2}{d^2} = 1$

Similarly gradient of tangent at $P = \frac{x_1}{c^2} \times \frac{d^2}{y_1}$

$$\begin{aligned} \therefore \text{Product of gradients} &= \frac{-x_1 b^2}{a^2 y_1} \times \frac{x_1 d^2}{c^2 y_1} \\ &= \frac{-x_1^2}{y_1^2} \times \frac{b^2 d^2}{a^2 c^2} \\ &= \frac{-a^2 c^2}{(a^2 - c^2)} \times \frac{(b^2 + d^2)}{b^2 d^2} \times \frac{b^2 d^2}{a^2 c^2} \\ &= \frac{-(b^2 + d^2)}{a^2 - c^2} \end{aligned}$$

from part (i)

But, conics have same foci so $ae = cE$

$$\begin{aligned} \text{and } b^2 &= a^2(1 - e^2) & \text{and } d^2 &= c^2(E^2 - 1) \\ b^2 &= a^2 - a^2 e^2 & d^2 &= c^2 E^2 - c^2 \end{aligned}$$

$$\begin{aligned} \therefore b^2 + d^2 &= a^2 - a^2 e^2 + c^2 E^2 - c^2 \\ &= a^2 - a^2 e^2 + a^2 e^2 - c^2 \\ &= a^2 - c^2 \end{aligned}$$

$$\begin{aligned}\therefore \text{Product of gradients} &= \frac{-(b^2 + d^2)}{a^2 - c^2} \\ &= \frac{-(a^2 - c^2)}{a^2 - c^2} \\ &= -1\end{aligned}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\alpha = \cos \theta + i \sin \theta$$

By de Moivre's theorem,

$$\alpha^k = \cos(k\theta) + i \sin(k\theta)$$

$$\alpha^{-k} = \cos(-k\theta) + i \sin(-k\theta)$$

$$= \cos(k\theta) - i \sin(k\theta)$$

$$\therefore \alpha^k + \alpha^{-k} = 2 \cos k\theta$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	3
• Sums the series, multiplies top and bottom by $(1 - \bar{\alpha})$ and attempts to expand, or equivalent merit	2
• Recognises that there are $2n + 1$ terms, or equivalent merit	1

Sample answer:

There are $(2n + 1)$ terms in the geometric series with first term α^{-n} and ratio α

$$\begin{aligned}
 C &= \frac{\alpha^{-n}(1 - \alpha^{2n+1})}{(1 - \alpha)} \\
 &= \frac{\alpha^{-n}(1 - \alpha^{2n+1})}{1 - \alpha} \times \left(\frac{1 - \bar{\alpha}}{1 - \bar{\alpha}} \right) \\
 &= \frac{\alpha^{-n} - \alpha^{-(n+1)} - \alpha^{(n+1)} + \alpha^n}{(1 - \alpha)(1 - \bar{\alpha})}, \quad \text{using } \bar{\alpha} = \frac{1}{\alpha} \\
 &= \frac{(\alpha^n + \alpha^{-n}) - (\alpha^{(n+1)} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})}
 \end{aligned}$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Correctly matches appropriate terms and attempts to use the result from part (i) , or equivalent merit	1

Sample answer:

Write C as

$$\begin{aligned}
 C &= 1 + (\alpha + \alpha^{-1}) + (\alpha^2 + \alpha^{-2}) + \cdots + (\alpha^n + \alpha^{-n}) \\
 &= 1 + 2\cos\theta + 2\cos 2\theta + \cdots + 2\cos(n\theta) \quad \text{by part (i)} \\
 &= 1 + 2(\cos\theta + \cos 2\theta + \cdots + \cos(n\theta))
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } C &= \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1 - \alpha)(1 - \bar{\alpha})} \\
 &= \frac{2\cos n\theta - 2\cos(n+1)\theta}{2 - (\alpha + \bar{\alpha})} \\
 &= \frac{2\cos n\theta - 2\cos(n+1)\theta}{2 - 2\cos\theta} \\
 &= \frac{\cos n\theta - \cos(n+1)\theta}{1 - \cos\theta}
 \end{aligned}$$

Question 16 (a) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Put $\theta = \frac{\pi}{n}$, then

$$1 + 2\left(\cos\frac{\pi}{n} + \cos\frac{2\pi}{n} + \cdots + \cos\frac{n\pi}{n}\right) = \frac{\cos\left(\frac{n\pi}{n}\right) - \cos(n+1)\frac{\pi}{n}}{1 - \cos\frac{\pi}{n}}$$

$$RHS = \frac{-1 - \cos\left(\pi + \frac{\pi}{n}\right)}{1 - \cos\frac{\pi}{n}}$$

$$= \frac{-1 + \cos\frac{\pi}{n}}{1 - \cos\frac{\pi}{n}}$$

$$= -1$$

$$\therefore LHS = 1 + 2\left(\cos\frac{\pi}{n} + \cos\frac{2\pi}{n} + \cdots + \cos\frac{n\pi}{n}\right)$$

$$= -1$$

$$\therefore \cos\frac{\pi}{n} + \cos\frac{2\pi}{n} + \cdots + \cos\left(\frac{n\pi}{n}\right) = -1$$

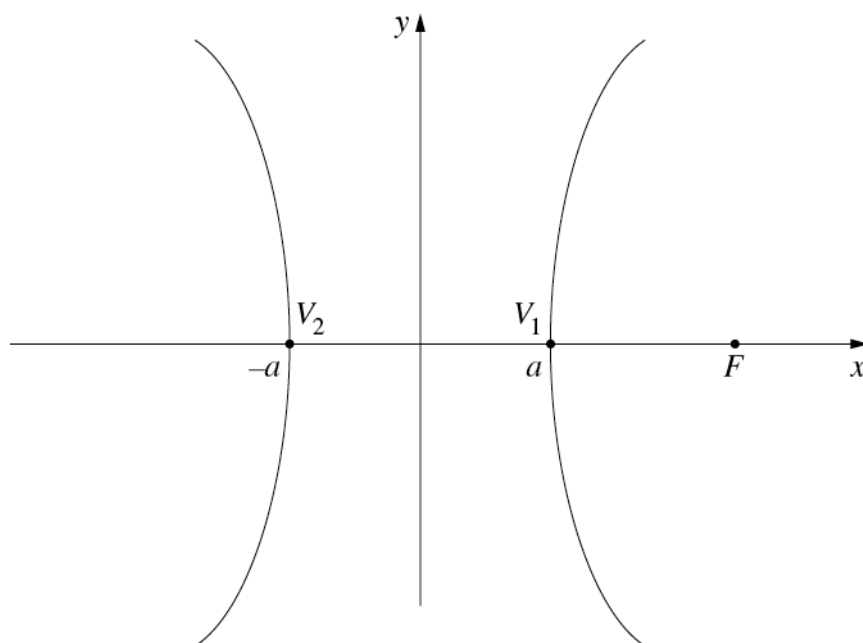
Which is independent of n .

Question 16 (b)

Criteria	Marks
• Provides correct solution	2
• Finds one correct value of a , or equivalent merit	1

Sample answer:

Since the eccentricity is 2 the conic is a hyperbola



The focus F is $(ae, 0) = (2a, 0)$

$$V_1F = 1, \text{ so } 2a - a = 1 \Rightarrow a = 1$$

$$FV_2 = 2a + a = 1 \Rightarrow a = \frac{1}{3}$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains correct count for one case, or equivalent merit	1

Sample answer:

A	C
B	D

Case 1: Tiles B and C have the same colour.
There are $(x - 1)$ ways of choosing tile D .

Case 2: Tiles B and C have different colours.
There are $(x - 2)$ choices of colour for tile C
and there are $(x - 2)$ choices of colour for tile D
giving $(x - 2)^2$ choices in total.

$\therefore$ There are $(x - 2)^2 + (x - 1) = x^2 - 3x + 3$ choices in total.

Question 16 (c) (ii)

Criteria	Marks
• Provides correct proof	2
• Demonstrates true for $n = 1$	1

Sample answer:

Let $P(n)$ be the given proposition.

$P(1)$ is true since this was given.

Let k be an integer for which $P(k)$ is true.

That is, there are $x(x - 1)(x^2 - 3x + 3)^{k-1}$ choices for the colours. Consider a $(k + 1) \times 2$ grid.

	A	C
	B	D

$\longleftarrow k \longrightarrow$

As in part (i) there are $(x^2 - 3x + 3)$ choices for the colours of tiles C and D .

$\therefore$ There are $x(x - 1)(x^2 - 3x + 3)^k$ choices for the colours.

$\therefore P(k + 1)$ is true, so $P(n)$ is true for all $n \geq 1$ by induction.

Question 16 (c) (iii)

Criteria	Marks
Provides correct solution	2
Finds the number of ways if not all colours need be used, or equivalent merit	1

Sample answer:

$$\text{Let } G(x) = x(x-1)(x^2-3x+3)^4$$

The number of ways to colour the grid with 3 colours, each used at least once, is

$$G(3) - 3 \times G(2) \text{ (there are 3 choices for which colour is left out)}$$

$$= 6 \times 3^4 - 3 \times 2 \times 1^4$$

$$= 480$$

2017 HSC Mathematics Extension 2

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	2.4	E3
2	1	3.4	E3
3	1	2.2	E3
4	1	1	E6
5	1	7.5	E4
6	1	7.5	E3
7	1	1.9	E6
8	1	1	E9
9	1	6.3.1	E5
10	1	8	E6

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	2.2	E3
11 (a) (ii)	2	2.2	E3
11 (b)	2	3.2	E3
11 (c)	2	2.5	E3
11 (d)	3	4.1	E8
11 (e)	2	5.1	E7
11 (f)	3	4.1	E8
12 (a) (i)	1	8	E6
12 (a) (ii)	1	8	E6
12 (a) (iii)	1	8	E6
12 (a) (iv)	1	8	E6
12 (a) (v)	1	1.5	E6
12 (b)	3	2.1	E3
12 (c)	3	4.1	E8
12 (d) (i)	2	7.2	E4
12 (d) (ii)	2	7.2	E4
13 (a)	1	8.3	E2
13 (b) (i)	2	7.5	E4
13 (b) (ii)	2	8.3	E4

Question	Marks	Content	Syllabus outcomes
13 (c)	4	6.2.2	E5
13 (d)	4	5.1	E7
13 (e)	2	2.3	E3
14 (a) (i)	1	7.6	E8
14 (a) (ii)	2	4.1	E8
14 (a) (iii)	1	4.1	E9
14 (b) (i)	1	8.1(2.9)	E2
14 (b) (ii)	1	8.1(2.9)	E2
14 (b) (iii)	3	8.1(2.9)	E2
14 (c) (i)	2	6.3.4	E5
14 (c) (ii)	2	6.3.3	E5
14 (c) (iii)	2	6.3.3, 6.3.4	E5
15 (a) (i)	1	4.1	E8
15 (a) (ii)	3	4.1	E8
15 (a) (iii)	1	4.1	E8
15 (b) (i)	2	1.8	E6
15 (b) (ii)	3	8	E6
15 (c) (i)	2	3.1, 3.2	E4
15 (c) (ii)	3	3.1, 3.2	E4
16 (a) (i)	1	2.4	E3
16 (a) (ii)	3	2.1, 8	E3
16 (a) (iii)	2	2.1, 8	E3
16 (a) (iv)	1	8	E3
16 (b)	2	3.2	E3
16 (c) (i)	2	8	E2
16 (c) (ii)	2	8.2	E2
16 (c) (iii)	2	8	E2